

Chemistry Ch 3.5: Isotopes

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Isotopes

Aise atoms jin ka **atomic number** ek ho magar **mass number** alag, yani **neutrons** alag.

Lafz Greek se hai: **isos** = ek jaisa, **tope** = jagah. Sab se pehle **Soddy** ne istemal kiya.

Isotopes **chemically** ek jaise hote hain, farq sirf **physical properties** ka hai.

Element	Us ke isotopes
Hydrogen	Protium H-1, koi neutron nahin, 99.99% · Deuterium H-2, 1 neutron, 0.0015% · Tritium H-3, 2 neutrons, radioactive
Carbon	C-12 (98.8%) · C-13 (1.1%) · C-14 (0.009%)
Chlorine	Cl-35 (75.77% , 17p + 18n) · Cl-37 (24.23% , 17p + 20n)
Uranium	U-234 (0.006%) · U-235 (0.72% , reactor aur bomb mein) · U-238 (99.27%)

Heavy water aur relative atomic mass

Heavy water (D₂O): melting point **3.81°C**, boiling point **101.2°C**, density **1.1044 g/cm³**. Aam paani se teenon zyada hain.

Weighted average: har isotope ka mass × us ki fiisad, phir jama, phir 100 se taqseem. Chlorine: $(35 \times 75.77 + 37 \times 24.23) \div 100 = \mathbf{35.48}$, jo table ke 35.453 ke bohat qareeb hai.

Isotopes ke istemal, yaad rakhne layak

Tibbi: I-131 **thyroid**, Na-24 **khoon ka bahao**, I-123 **dimagh ki tasveer**, Co-60 **cancer ka ilaj**.

Tehqeeq: C-14 photosynthesis ka raasta dekhne ke liye, S-35 molecule ki banawat ke liye.

Carbon dating: C-14 se carbon wali cheezon ki **umar** maloom ki jati hai: chattanein, mitti, mummies.

Quiz

1. Define **isotopes**. Name the three isotopes of hydrogen with their neutrons, and give the two isotopes of chlorine with their percentages.
2. Calculate the relative atomic mass of chlorine from Cl-35 (75.77%) and Cl-37 (24.23%). Then give **four** uses of isotopes, naming the isotope in each case.